NeuroHack Proposed Judging Rubric

Working draft for a clearer judge-facing scoring rubric

Purpose. This draft is meant to help judges score more consistently across technical and non-technical submissions. To make the score bands easier to use in real time, all seven categories are set to 10 marks each (70 total marks total). This should be treated as a proposed internal judging guide unless the organizing clubs choose to formally adopt it.

Suggested scoring norm: use 10 sparingly for truly outstanding work in that category; 8–9 for strong work with minor gaps; 5–7 for promising but still limited work. Scores below 5 can be used when a category is largely missing or seriously weak.

Category Weights

Category	Marks	Why it matters
1. Relevance & Problem Understanding	10	Ensures the project targets a meaningful attention, focus, or cognitive-performance problem.
2. Neuroscience & Conceptual Depth	10	Rewards scientific rigor and accurate use of neuroscience.
3. Interdisciplinary Integration	10	Values genuine synthesis across fields, not token inclusion.
4. Innovation & Creativity	10	Recognizes originality and distinctiveness of the approach.
5. Execution & Technical Quality	10	Assesses quality of build, logic, or validation.
6. Real-World Impact, Feasibility & Scalability	10	Prioritizes plausible adoption, testing, and real-world value.
7. Presentation & Communication	10	Helps judges reward clarity, structure, and overall communication quality.

Detailed Performance Anchors

Category	Marks	5–7 marks Developing / partial	8–9 marks Strong	10 marks Exceptional
1. Relevance & Problem Understanding	10	Problem is loosely connected to attention, focus, cognitive performance, or cognitive wellness. The target user or pain point is vague, oversimplified, or assumed rather than demonstrated. Real-world stakes are mentioned, but the framing does not yet show strong understanding of who is affected, why the issue matters, or what meaningful success would look like.	Project addresses a clear and relevant cognitive-wellness problem with a defined user group or use case. The team shows a good grasp of the practical challenge and explains why it matters. Problem framing is coherent, though some assumptions, edge cases, or contextual factors may still need refinement.	Problem selection is highly relevant, sharply framed, and clearly tied to a real cognitive or behavioral challenge. The team demonstrates nuanced understanding of the user, context, and constraints, and defines a compelling problem statement that makes the project's purpose immediately clear.
2. Neuroscience & Conceptual Depth	10	Neuroscience language is present, but largely superficial, generic, or decorative. Mechanisms such as attention networks, cognitive control, fatigue, distraction, reward, or learning are referenced without being meaningfully integrated into the design. Limited evidence of grounding in literature, theory, or established brain-behavior principles.	Project uses relevant neuroscience concepts appropriately and connects them to the solution in a credible way. Scientific ideas are more than buzzwords and generally support the design choices. Evidence base is present, though the translation from theory to intervention, model, or interface could be more rigorous or specific.	Neuroscience is meaningfully embedded in the project logic. The team demonstrates strong conceptual accuracy, uses mechanism-level reasoning where appropriate, and grounds the approach in relevant theory, literature, or established principles of brain and behavior. Scientific framing strengthens the solution rather than merely describing it.
3. Interdisciplinary Integration	10	Multiple disciplines may be mentioned, but they operate in parallel or appear bolted on. Connections among neuroscience, computation, BCI/neurotechnology, behavioral science, design, or user research are weak, incomplete, or mostly symbolic.	Project integrates at least two disciplines in a functional way. The components are clearly related and support the overall concept, though the synthesis may still feel uneven or underdeveloped in places. Some elements are integrated well, while others remain more additive than truly unified.	Project shows genuine interdisciplinary synthesis. Methods, theory, and implementation from different domains are deliberately combined into a coherent whole, with each discipline clearly strengthening the others. The result feels unified rather than assembled from separate pieces.
4. Innovation & Creativity	10	Idea is familiar, incremental, or derivative, with limited novelty in framing, implementation, or user experience. Any creative elements do not substantially change how the problem is addressed.	Project shows a recognizable original angle, useful recombination of existing ideas, or thoughtful creative execution. While not fully breakthrough-level, the concept stands out from generic solutions and demonstrates purposeful design choices.	Project is notably original, imaginative, and strategically inventive. Novelty is evident in the core concept, the way tools or systems are used, or how the problem is reframed. Creativity materially improves the project's distinctiveness and value.
5. Execution & Technical Quality	10	Prototype, workflow, or concept support is incomplete, fragile, poorly justified, or difficult to follow. Implementation may contain major logic gaps, weak technical choices, or insufficient evidence that the	Project is reasonably well executed, with a working prototype, clear architecture, or credible non-technical support materials. Core functionality or logic is understandable and generally sound, though polish,	Execution is polished, logically sound, and professionally assembled. Technical implementation or concept validation is clear and credible, with strong alignment between idea and build. The team demonstrates

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		solution can work as presented.	robustness, testing, or implementation depth may be limited.	command of the practical details needed to support confidence in the solution.
6. Real-World Impact, Feasibility & Scalability	10	Potential impact is asserted but not well substantiated. Use case, implementation path, testing environment, or target population is underdeveloped. Significant barriers to adoption, ethics, usability, cost, or scale are not meaningfully addressed.	Project has a plausible use case and a reasonable path to pilot, test, or iterate in the real world. Impact is understandable, and the team acknowledges some practical constraints. Scalability or deployment considerations are present but may still need depth.	Project demonstrates strong real-world promise with a clearly defined use case, credible path to testing or adoption, and thoughtful attention to feasibility constraints. The team addresses next steps, stakeholder fit, and scalability in a way that makes future development feel realistic.
7. Presentation & Communication	10	Explanation is difficult to follow, overly vague, or insufficiently professional. Important details are missing, visuals do not clarify the idea, or the project story does not clearly connect problem, solution, and impact.	Presentation is clear, organized, and mostly effective. Judges can understand the project, rationale, and intended use. Communication supports evaluation well, though narrative flow, visual economy, or persuasive clarity could still improve.	Presentation is polished, concise, and compelling. The team communicates the problem, scientific rationale, solution, and impact with clarity and professionalism. Visuals and verbal or written explanation actively strengthen understanding and judge confidence.

Optional Judge Feedback Prompts

These prompts can help judges leave useful written feedback alongside numeric marks. Judges do not need to answer every prompt. The goal is simply to make feedback more specific and more helpful for teams.

Feedback area	Suggested prompt for judges
Strongest aspect	What is the single strongest element of this project, and why does it stand out?
Most important improvement	What one improvement would most increase the project’s overall quality or readiness?
Scientific confidence	How convincing is the neuroscience rationale, and what additional evidence would strengthen it?
Feasibility note	What is the biggest practical obstacle to testing, adoption, or scaling?
User-centered note	How well does the project reflect the needs of the intended user or population?
Optional judge comment	Any domain-specific insight, caution, or opportunity the team should hear from an expert reviewer?

Suggested deliberation norm: if judges differ a lot, first compare evidence for neuroscience grounding, feasibility, and user value before debating style or polish.